

BAYSTEN ONBOARDING OPTIMIZATION

*Reducing User Drop-Off Through AI-Powered Resume Parsing
and a Lower-Friction Profile Experience*

Prepared for: Baysten

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Ross School of Business Consulting Studio | April 2026

Executive Summary

32 mins average manual completion time	13% profile completion rate at baseline	134 resumes tested in validation corpus	95.3% average field-level accuracy	109/134 resumes scored at 100% accuracy	< 2 min target completion with resume upload
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Baysten operates at the intersection of candidate discovery, recruiter efficiency, and AI-powered matching. Its search engine can only perform as well as the candidate data it receives. Over the course of this engagement, the team found that Baysten's most immediate and most solvable growth constraint is not the ranking engine itself. It is the gap between app download and profile completion. Candidates are being asked to provide a high volume of structured information before they experience meaningful value, and that burden creates drop-off at the exact moment Baysten needs trust, momentum, and data quality most.

The project began as a broader exploration of search quality, embedding accuracy, and platform performance. As the team worked through Baysten's technical environment, sponsor conversations, and faculty feedback, the core problem became clearer: search relevance, recruiter visibility, and long-term monetization all depend on complete, structured candidate profiles. In Baysten's current flow, profile creation is largely manual, section-heavy, and cognitively demanding. The team documented an average completion time of more than 30 minutes and a post-registration profile completion rate of just 13%.

In response, the team reframed the engagement around onboarding optimization and built a working proof of concept: an LLM-based resume parser, an editable review layer, and a redesigned single-page onboarding flow. The prototype was evaluated on a corpus of 134 real resumes. Using a manually reviewed ground-truth file, the system achieved 95.3% average field-level accuracy, well above the stated 85% target in the PRD. 109 of 134 tested resumes scored at 100% across all evaluated fields, and 131 of 134 scored at or above 80%.

The team recommends a three-part implementation path: (1) integrate the resume parser into the current onboarding architecture while preserving an editable review screen; (2) redesign profile creation into a continuous page that makes requirements visible and minimizes redundant navigation; and (3) separate first-session essentials from later-session enrichment. The expected outcome is a faster, more credible, and more scalable onboarding flow that improves completion rates and strengthens the underlying search product simultaneously.

Key Decision for Baysten

- Adopt resume upload and autofill as a core onboarding capability, not a nice-to-have enhancement.
- Treat profile completion rate as a business-critical product metric because it directly affects search quality and monetization potential.
- Use the delivered prototype as the foundation for staged implementation, beginning with standard-format resumes and an editable verification layer.

1. Project Scope and Business Context

Baysten is building an AI-enabled professional networking and recruiting platform with a strong early focus on healthcare talent and cross-border matching. The company's search approach combines structured taxonomy fields, semantic embeddings, and ranking logic to help recruiters identify relevant candidates efficiently. In Baysten's model, profile completeness is not a cosmetic issue. It is an operating requirement. If critical fields are missing, candidates become harder to rank, easier to miss, and less useful to both recruiters and Baysten's own AI systems.

Sponsor discussions and project materials showed that Baysten's long-term business model depends on user trust, recruiter outcomes, and scalable domain expansion. The company currently monetizes through premium usage and recruiting success fees, with future value also tied to the quality of the talent graph it is building. That context shaped the team's central conclusion: improving onboarding is not separate from improving Baysten's business. It is one of the fastest ways to increase the quantity and quality of searchable talent on the platform.

The signed Letter of Engagement narrowed the final project scope to onboarding friction, user drop-off, and testable product recommendations. The team was not asked to redesign recruiter search, rebuild ranking algorithms, or deploy production code directly into Baysten's environment. Instead, the task was to diagnose the highest-friction parts of the onboarding journey, validate practical solutions, and deliver an implementation-ready proof of concept supported by research, logic, and technical documentation.

2. Research and Methodology

The project evolved through four connected phases rather than a single linear workstream. The methodology was designed to move from context-setting to evidence-gathering to prototype development to validation, with sponsor feedback woven into each stage.

Phase 1: Project Entry and Alignment

Early meetings focused on Baysten's company context, technical architecture, and the original project scope, which centered on search quality, embeddings, and user experience. These conversations were critical in establishing what the platform needed structurally and where a consulting team's time would have the most impact. The team reviewed Baysten's MongoDB and Weaviate architecture, understood the profile schema, and mapped how profile fields connect to search indexing and recruiter-facing outcomes.

Phase 2: Current Flow Diagnosis

The team assessed the current onboarding experience directly, walking through the app end-to-end and documenting every field, section, and step in the user journey. This included reviewing both required and optional fields, estimating realistic time burden for a first-time user, and identifying where the cognitive load concentrated most. The team documented an average manual completion time of over 30 minutes and a baseline profile completion rate of 13% at the post-OTP stage.

Phase 3: Prototype Development

The team translated findings into a working proof of concept. The prototype included a React/Vite frontend for resume upload, an Express.js backend proxy managing API authentication, an LLM-based parsing layer using Claude's structured JSON extraction capability, an editable review screen, and a redesigned onboarding flow. The PRD and roadmap documents defined the target data schema and scope boundaries.

Phase 4: Accuracy Testing and Validation

The team built a systematic batch testing framework to validate parser performance. 134 real resumes were run through the parser, and extracted output was compared against a manually reviewed ground-truth CSV. The scoring logic evaluated eight fields selected for their direct impact on search usefulness in the first session: name, email, phone, location, current title, current company, school, and degree.

Method	Purpose	What It Contributed	Limitation
Sponsor interviews and weekly meetings	Clarify Baysten context and constraints	Connected onboarding decisions to search quality, business model, and implementation feasibility	<i>Sponsor availability limited depth of some technical discussions</i>
Current flow walkthroughs	Understand lived onboarding burden	Surfaced manual data entry, redundant fields, and the true time cost of profile creation	<i>Team members as proxies for real users; not a formal user study</i>
PRD and roadmap development	Translate insight into product design	Defined MVP scope, data schema, validation logic, and technical architecture	<i>Scope evolved during the project; some early PRD assumptions revised</i>
Prototype build	Test feasibility	Demonstrated that resume upload, extraction, and user review work in a Baysten-oriented flow	<i>Not integrated into Baysten's production environment</i>
Ground-truth accuracy testing	Quantify performance	Provided a defensible evidence base for deployment readiness and improvement priorities	<i>Ground truth covered 8 fields; many more fields are relevant for full profile completion</i>

3. How Metrics Were Selected

One of the faculty's specific feedback items was that the team should explain how metrics were chosen, not just what they were. That reasoning matters because the choice of what to measure shapes everything that follows: how the prototype was designed, which fields were prioritized in testing, what success looks like for Baysten after implementation, and how the team justified its recommendations. The selection process involved three criteria.

Criterion 1: Direct Connection to Search Quality

The first filter was whether a field or metric directly influences a candidate's visibility in Baysten's search engine. Baysten's AI system uses structured profile data stored in MongoDB and vector embeddings in Weaviate to match candidates with recruiter queries. Fields that are empty cannot contribute to search ranking. That made name, current title, current company, school, degree, and skills the most urgent targets for extraction accuracy, because their absence most immediately damages a candidate's discoverability.

Criterion 2: User Effort Reduction Potential

The second filter was whether improving extraction for a given field would noticeably reduce the burden on the user. Contact fields like name, email, phone, and location are high-effort to type but low-effort for the parser to extract from a standard resume. Work history and education are the sections most responsible for abandonment because they require multiple structured entries. These fields therefore became primary accuracy targets because improving extraction here has the largest practical effect on perceived onboarding speed.

Criterion 3: Measurability Post-Launch

The third filter was whether the metric could be tracked continuously in production, not just in a lab setting. Metrics that require manual review are useful once but do not scale. The team therefore prioritized metrics that could be instrumented automatically: profile completion rate via backend event logging, field acceptance rate via UI interaction events, time to searchable profile via Weaviate write timestamps, and upload-to-submit conversion via funnel tracking. Parser accuracy in production is derived from field acceptance rate as a proxy rather than requiring continuous manual ground-truth scoring.

Figure 1: Metric Selection Framework and Measurement Approach

Metric	Why This Metric	How to Measure	Tier
Profile Completion Rate	Most direct signal that onboarding friction has decreased – measures % of users who reach a searchable profile state	Tracked in Baysten backend on account creation events; baseline is current 13% post-registration rate	PRIMARY
Upload-to-Submit Conversion	Measures whether resume upload actually converts to a completed profile, not just an attempted one	Event logged when user clicks upload vs. when they hit 'confirm and save'; delta reveals trust/UX gaps in the review step	PRIMARY
Field Acceptance Rate	Shows how often users accept autofilled data without editing – a proxy for extraction quality in real usage beyond lab testing	Log event per field: 'kept', 'edited', or 'deleted'; low acceptance on a specific field flags a parsing or normalization issue	TECHNICAL
Time to Searchable Profile	Measures user effort more honestly than session time alone – counts only until the profile becomes search-indexable	Timestamp from account creation to first successful embedding write in Weaviate; compare pre-parser vs. post-parser cohorts	PRIMARY
Section Completion Rate	Reveals where remaining friction lives after the redesign; critical for identifying which optional fields actually get filled	Track per-section fill rates in the continuous profile builder; section-level drop-off map guides progressive disclosure decisions	SECONDARY
Parser Accuracy (ongoing)	Ensures production quality does not degrade as new resume formats, languages, and edge cases enter the system	Monthly ground-truth batch scoring against a growing benchmark corpus; flag fields where edit rate in production diverges from lab accuracy	TECHNICAL
Net Promoter Score (onboarding)	Captures subjective candidate experience beyond conversion metrics	Single-question NPS prompt displayed 24 hours after profile completion; track alongside completion rate to separate effort from satisfaction	SECONDARY

4. Analysis: Why Users Are Dropping Off

The central insight of the project is simple but important. Baysten currently asks users to do too much before they understand the payoff. In the existing flow, candidates register, verify their contact information, and then face a long sequence of fields across personal information, experience, salary, education, contact, interests, additional information, and supporting documents. Many of those fields may matter eventually, but they do not all need to be asked at the same time.

Figure 2: Current Onboarding Flow (with identified drop-off point)

Step	Stage	What Users Must Do	Status
01	User Downloads App	Entry point	
02	Creates Account	Email + password + OTP	
03	Enters Personal Info	Name, DOB, pronouns, citizenship	
04	Adds Work History	<i>Each role entered separately (highest drop-off)</i>	CRITICAL DROP-OFF
05	Adds Education	School, degree, dates – manual entry	
06	Sets Preferences	Salary, availability, employment type	
07	Uploads Documents	Supporting materials	
08	Submits Profile	Only 13% of users reach this step	13% arrive

From a product perspective, three issues combine to create friction. The first is effort volume: users must type a significant amount of information manually, often repeating content that already exists on their resume or LinkedIn profile. The second is cognitive load: several fields are either ambiguous, low-value in the initial decision moment, or difficult to answer quickly on mobile. The third is sequencing: users are asked for depth before value, which means candidates are required to demonstrate full commitment to the platform before they have experienced its core matching functionality.

Manual Entry Is the Primary Barrier

The current onboarding process requires manual population of core profile data, education entries, experience entries, contact details, and additional sections. Education and experience must be added one item at a time, which increases perceived effort and breaks momentum. The team's walkthrough documented an average completion time of over 30 minutes, with the longest burden concentrated in work history. This matters because the drop-off is not random. It is structurally built into the design of the journey.

Some Fields Create Friction Without Delivering First-Session Value

The team found that several fields in the original flow are either not critical to initial search usefulness or can be collected later. Date of birth, multiple social media links, highly specific salary details, and some additional-information prompts all increase data-entry load without clearly improving a recruiter's ability to identify a viable candidate in the first pass. For an early-stage platform still earning trust, that gap matters. Fields like availability, employment type, and willingness to relocate are more defensible because they influence recruiter decision-making more directly and can often be inferred or provided quickly.

Incomplete Profiles Weaken the Search Product Itself

Baysten's search system depends on structured fields being available in MongoDB and on meaningful embeddings being generated for Weaviate. Missing fields reduce ranking quality, hide candidates from recruiter searches, and create extra manual work for Baysten staff. Profile incompleteness does not just hurt the candidate journey. It compromises the product engine Baysten is trying to scale. This is where onboarding becomes a revenue issue, not merely a UX one.

5. The Proposed Solution

The solution the team is recommending is not a single feature. It is a coordinated redesign of how Baysten captures candidate data. The first layer is an LLM-based resume parser that converts uploaded documents into structured fields. The second layer is an editable review experience that lets candidates verify and correct extracted information before submission. The third layer is a redesigned onboarding flow that presents upload, review, and manual completion in one continuous experience.

Figure 3: Proposed Onboarding Flow (streamlined with AI autofill)

Step	Stage	What Happens
01	User Downloads App	Entry point – same as before
02	Creates Account	Email + password only – streamlined
03	Uploads Resume	PDF or DOCX – one action replaces five manual steps
04	AI Parses Resume	LLM extracts all key fields in seconds
05	User Reviews Profile	Editable preview – confirm or correct extracted data
06	Selects Skills	Auto-suggested from specialty field; add/remove freely
07	Profile Complete	Searchable and ready – target: under 2 minutes total

Technical Architecture

The implemented proof of concept follows a six-stage pipeline from resume upload to profile storage. Each stage was designed with Baysten's existing infrastructure in mind: MongoDB for structured candidate profile storage, Weaviate for vector embedding generation, and the Anthropic API for LLM-based field extraction.

Figure 4: Technical Pipeline – From Resume Upload to Searchable Profile

Phase	Component	Technical Detail
INPUT	Resume Upload	User uploads PDF or DOCX file via React/Vite frontend. File size capped at 5 MB. DOCX converted to plain text using mammoth.js. PDF content extracted via pdfplumber or base64 encoding.
EXTRACT	Text Extraction	Raw resume text is cleaned and passed to the backend Express.js proxy. The proxy manages API key security and structures the LLM prompt with explicit JSON output instructions and field-level definitions.
PARSE	LLM Parsing (Claude API)	Claude Sonnet processes the resume text and returns a structured JSON object. The prompt specifies eight core fields: name, email, phone, location, current title, current company, school, and degree. A system-level instruction enforces output format and prevents hallucination.
MAP	Schema Mapping	Extracted JSON is mapped to Baysten's profile schema fields. Field normalization handles formatting differences (e.g., phone number formats, location abbreviations). Confidence flags mark fields with ambiguous extraction.
REVIEW	Editable User Interface	All extracted fields are displayed in an editable preview. Users can confirm, correct, or add missing information before submission. The interface tracks which fields were changed (field acceptance rate) as a signal for ongoing parser improvement.
STORE	Profile Storage	Confirmed profile data is formatted for MongoDB document storage. Embeddings are generated for Weaviate vector search indexing. The profile becomes immediately searchable for recruiters.

LLM Prompt Design and Why It Matters

The quality of structured extraction from unstructured resume text depends heavily on how the prompt is designed. The team's approach defined each target field explicitly in the system prompt, specified the expected output format as a JSON object with named keys, included fallback instructions for fields that may be absent or ambiguous, and explicitly prohibited hallucination by instructing the model to return null for fields it cannot find rather than inferring plausible values. This last element is critical for a trust-sensitive application: an inaccurate name or degree extracted with false confidence is worse than a blank field the user can fill in manually.

The prompt also included field-specific normalization guidance. For location fields, the model was instructed to output a standardized city-country format rather than raw address strings. For degree fields, the model was given a list of common degree abbreviations to resolve against. For employment dates, it was instructed to output ISO-format date strings rather than resume-style notation. These normalization instructions directly addressed the most common causes of downstream schema mapping failures.

6. Results and Validation

The strongest quantitative evidence from the project comes from parser validation. The team tested the prototype on 134 real resumes and compared parser output against a manually reviewed ground-truth CSV file. Across the eight scored fields, the parser achieved 95.3% average field-level accuracy, exceeding the PRD target of 85%. The median accuracy score was 100%, meaning the typical standard-format resume parsed completely cleanly.

Metric	Result	Interpretation
Resumes tested	134	Large enough to identify real edge cases rather than only ideal examples
Average field accuracy	95.3%	Prototype materially exceeded the stated 85% success target in the PRD
Median accuracy	100%	Typical standard-format resume parsed completely cleanly across all scored fields
Resumes at 100% accuracy	109 of 134 (81.3%)	The large majority of tested resumes required little to no correction on scored fields
Resumes at or above 80%	131 of 134 (97.8%)	Performance is strong across the tested corpus; 3 outliers identify edge case categories for improvement
Weakest field	Degree (84.8%)	Inconsistent degree naming conventions, abbreviations, and multilingual variations drive most errors
Strongest fields	Name, Email, Title, Company, School (100%)	High-frequency, well-structured fields parse with near-perfect reliability

Field-Level Accuracy Results

Figure 5: Field-Level Extraction Accuracy (134-resume corpus)

Field	Extraction Accuracy (visual)	Score	Status
Name		100%	Excellent
Email		100%	Excellent
Current Title		100%	Excellent
Current Company		100%	Excellent
School		100%	Excellent
Phone		99.2%	Good
Location		94%	Good
Degree		84.8%	Improve

Performance varied by field in meaningful ways. The parser performed exceptionally well on name, email, current title, current company, and school extraction because these fields are typically expressed in standardized, prominent positions on standard resumes. Degree extraction was notably weaker, which is unsurprising given the range of degree naming conventions, abbreviations, and formatting patterns across resumes. That finding tells Baysten exactly where additional normalization and prompt refinement are likely to generate the most immediate gains.

Resume Format Variety Tested

Figure 6: Resume Format Variety and Parsing Performance

Resume Type	Format	Language	Primary Parsing Challenge	Avg Accuracy
Standard Chronological	PDF / DOCX	English	Minimal – clean section headers and consistent date formatting	100%
Functional / Skills-First	PDF	English	Skills listed before experience; title may appear mid-document	97%
Academic / Research CV	PDF	English	Publications, committees, and multiple institutional affiliations complicate experience parsing	93%
Multi-column Designed	PDF	English	Two-column layouts disrupt text extraction order; sidebar fields often misaligned	88%
International / Multilingual	PDF / DOCX	Mixed	Mixed script fields; address and phone formats differ by country; degree names in non-English	85%
Healthcare / Clinical	PDF	English	Certifications and clinical specialties use domain-specific abbreviations (RN, BSN, ICU)	91%
Freelance / Portfolio	DOCX	English	No single employer; project-based work without clear title or company field	84%
Scanned / Image-Based	PDF (image)	English	No embedded text; requires OCR; out of scope for MVP but identified for Phase 2	N/A*

The corpus included chronological, functional, academic, multi-column designed, healthcare-specific, freelance, and multilingual resume formats. Standard chronological resumes and healthcare-specific formats performed best. Multi-column designed resumes and multilingual formats showed more variability due to text extraction challenges. Image-based scanned PDFs were identified as an important future scope item requiring OCR integration, currently out of scope for the MVP.

7. Recommendations

Based on the analysis and prototype validation, the team recommends that Baysten take three actions in sequence. These recommendations are intentionally practical. They do not depend on a full platform rebuild, and they are designed to create measurable improvement in the part of the product where Baysten can move fastest.

Recommendation 1: Deploy Resume Upload and Editable Autofill

Baysten should treat resume parsing as a core onboarding capability, not an enhancement. The technical proof of concept is already strong enough to justify a staged deployment for standard-format resumes, especially because the workflow keeps users in control through an editable review screen. This is the single highest-leverage change because it removes the largest source of manual effort. Baysten should launch with standard text-readable resumes first, instrument the flow carefully, and use real correction behavior to drive later model improvements.

Recommendation 2: Redesign Onboarding into One Continuous Profile Experience

The current multi-section experience increases abandonment by turning onboarding into a series of stops. Baysten should move toward a single continuous profile builder where upload, preview, skills, and additional edits appear in one visible flow. This structure keeps users oriented, reduces repeated navigation, and lowers perceived effort because the task feels bounded. It also makes it easier for Baysten to show which fields are required, which are optional, and how far the user is from a searchable profile.

Recommendation 3: Rationalize Required Versus Optional Fields

Baysten should reduce the number of fields that matter in the first session. The minimum viable profile should focus on information that directly influences search visibility and recruiter utility. Fields like contact information, core skills, current role, education, availability, employment type, and relocation preference are more defensible than date of birth, multiple social channels, or high-effort details that can be added later.

Recommended Minimum Viable Profile

- **Identity and contact:** name, email, phone, location
- **Professional snapshot:** current title, current company, education, key skills (minimum 5)
- **Search-enabling preferences:** availability, desired employment type, willingness to relocate
- **Review layer:** user confirmation and editing before submission
- **Deferred enrichment:** additional certifications, long-form experience details, hobbies, supplementary documents

8. Implementation Roadmap

Because Baysten already has working prototype code and clear product requirements, implementation can be staged rather than speculative. The most important principle is to launch narrowly and learn quickly. A limited first release focused on standard resume formats and an editable review step gives Baysten real usage data without creating unnecessary operational risk.

Figure 7: Implementation Roadmap

Timeline	Actions
Phase 1 <i>0–2 weeks</i> Controlled Deployment	• Integrate resume upload into existing onboarding • Preserve editable review screen • Track upload-to-completion conversion • Monitor field acceptance rates from day one
Phase 2 <i>2–4 weeks</i> Flow Redesign	• Consolidate multi-step flow into single continuous page • Make required vs. optional fields explicitly visible • Add progress indicator to reduce perceived effort • A/B test against current multi-step layout
Phase 3 <i>4–8 weeks</i> Field Rationalization	• Remove or defer low-value first-session fields (DOB, redundant social links) • Implement minimum viable profile requirement set • Show optional fields progressively after initial submission • Measure completion rate impact of each field change
Phase 4 <i>Ongoing</i> Model Refinement	• Use real field edit rates to improve extraction prompts • Expand corpus for multilingual and designed resume formats • Add OCR support for image-based PDFs • Set monthly accuracy benchmarks against growing ground-truth set

9. Risks, Trade-offs, and How to Manage Them

The main trade-off in resume parsing is speed versus certainty. Baysten can dramatically reduce effort, but it cannot assume perfect extraction in every case. That is why the editable review layer is essential. The goal is not zero-error automation. The goal is to make correction lighter than creation. There are also known technical risks around layout-heavy resumes, scanned documents, multilingual content, and inconsistent degree naming conventions. None of these issues invalidate the approach. They simply argue for staged rollout, careful instrumentation, and ongoing iteration.

Risk	Likely Impact	Mitigation
Low-confidence extractions (e.g., degree names)	User sees inaccurate data in review screen; may reduce trust in the parser	Flag low-confidence fields visually in the UI; prompt user to verify specifically those fields
Multi-column or designed resume layouts	Text extraction order breaks; fields may be misattributed or merged	Expand preprocessing to handle two-column PDFs; use layout-aware extraction for high-frequency format types
Multilingual and mixed-script resumes	Field boundaries harder to identify; transliteration may affect schema matching	Add language detection before parsing; include multilingual degree and title normalization in prompt
Image-based scanned PDFs	No embedded text; extraction fails entirely	Detect image-only PDFs early and prompt user to upload a text-based version; defer OCR to Phase 2
Parser accuracy drift over time	Real-world resumes diverge from test corpus; accuracy degrades without feedback loop	Instrument field acceptance rates; set monthly benchmark runs; retrain prompts when edit rates spike on specific fields

10. Conclusion

This project started in a technically ambiguous place and ended in a business-relevant one. Baysten initially asked broad questions about search quality, embeddings, and platform performance. What the work ultimately showed is that Baysten's search product cannot reach its potential without solving the problem that comes before search: getting candidates to complete strong profiles in the first place. That is the real bottleneck. It is also one of the few bottlenecks Baysten can address quickly.

The team's final recommendation is direct. Baysten should implement resume upload with editable autofill, redesign onboarding into a continuous and lower-friction flow, and move to a minimum viable profile model that captures what matters first. The prototype and the validation results indicate that this is not just a plausible direction. It is an actionable one. The parser already exceeds its stated accuracy target across a diverse corpus of 134 real resumes. The onboarding redesign concept has been validated with faculty and sponsor feedback. The implementation path is staged, practical, and does not require a platform rebuild.

What made this project genuinely interesting was the pivot itself. Coming in, the team expected to spend most of its time on algorithms and embeddings. What actually mattered was a much more human problem: the experience of filling out a form at 30 minutes of cognitive effort, on a mobile device, before you have any reason to trust the platform. Resume parsing is not glamorous. But fixing the onboarding gap may well be the most consequential thing Baysten can do right now, because every other product investment depends on having good profiles to work with. If Baysten follows through on these recommendations, the likely result is not only a better onboarding experience, but also a stronger candidate database, a more useful search engine, and a more credible product for the market it is trying to serve.

Technical Appendix

Appendix A: Prototype Scope Delivered

- Resume upload interface accepting PDF and DOCX formats (image-based PDFs identified for Phase 2 scope)
- LLM-based extraction into structured JSON using Claude Sonnet via Anthropic API
- Editable review screen for extracted candidate data with field-level confirmation
- Single-page onboarding concept combining upload, review, and manual refinement
- Ground-truth accuracy scoring framework using 134 real resumes against a manually reviewed CSV
- Batch testing infrastructure (batch_test.js and score_accuracy.js) for ongoing parser evaluation
- Documentation covering architecture, schema mapping, prompt design, and future extension paths

Appendix B: Core Fields Targeted in the Roadmap

Category	Fields	Notes
Contact	Name, email, phone, location, selected links	Highest-confidence extraction; critical for profile identity and recruiter contact
Professional Experience	Company, title, dates, city, responsibilities, specialty	Title and company parse cleanly; responsibilities require truncation logic for verbose entries
Education	School, degree, field of study, level, dates, country	School parses cleanly; degree and field require normalization for abbreviations and non-English conventions
Skills and Languages	Extracted skill list, spoken languages	Auto-suggested in UI; user can add, edit, or remove from suggested set
Search Preferences	Availability, employment type, relocation willingness	Not parsed from resume; collected through quick-select UI during onboarding review step
Deferred Enrichment	Certifications, supporting documents, team leadership, hobbies	Collected after initial profile submission to avoid first-session overload

Appendix C: Suggested Post-Launch Metrics (detailed)

Metric	Why This Metric	How to Measure	Tier
Profile Completion Rate	Most direct signal that onboarding friction has decreased – measures % of users who reach a searchable profile state	Tracked in Baysten backend on account creation events; baseline is current 13% post-registration rate	PRIMARY
Upload-to-Submit Conversion	Measures whether resume upload actually converts to a completed profile, not just an attempted one	Event logged when user clicks upload vs. when they hit 'confirm and save'; delta reveals trust/UX gaps in the review step	PRIMARY
Field Acceptance Rate	Shows how often users accept autofilled data without editing – a proxy for extraction quality in real usage beyond lab testing	Log event per field: 'kept', 'edited', or 'deleted'; low acceptance on a specific field flags a parsing or normalization issue	TECHNICAL
Time to Searchable Profile	Measures user effort more honestly than session time alone – counts only until the profile becomes search-indexable	Timestamp from account creation to first successful embedding write in Weaviate; compare pre-parser vs. post-parser cohorts	PRIMARY
Section Completion Rate	Reveals where remaining friction lives after the redesign; critical for identifying which optional fields actually get filled	Track per-section fill rates in the continuous profile builder; section-level drop-off map guides progressive disclosure decisions	SECONDARY
Parser Accuracy (ongoing)	Ensures production quality does not degrade as new resume formats, languages, and edge cases enter the system	Monthly ground-truth batch scoring against a growing benchmark corpus; flag fields where edit rate in production diverges from lab accuracy	TECHNICAL
Net Promoter Score (onboarding)	Captures subjective candidate experience beyond conversion metrics	Single-question NPS prompt displayed 24 hours after profile completion; track alongside completion rate to separate effort from satisfaction	SECONDARY

Appendix D: Accuracy Testing Methodology

The batch testing framework consisted of two JavaScript modules. The first, `batch_test.js`, iterated over a directory of 134 resume files, called the parser pipeline for each, and saved structured JSON output to a results directory. The second, `score_accuracy.js`, loaded the results alongside a manually reviewed ground-truth CSV and computed field-level accuracy scores using exact string matching with normalization (lowercased, trimmed, punctuation-stripped) as well as a partial-match threshold for longer fields like location and degree. Fields scored as empty in the ground truth were excluded from the denominator to avoid penalizing the parser for fields genuinely absent from the resume.

The 134-resume corpus was constructed to represent realistic variation: it included resumes from international candidates, recent graduates, mid-career professionals, and healthcare specialists. Edge case resumes covering multilingual content, degree-in-progress notation, freelance work history, and CJK-script names were manually curated and added to the corpus specifically to stress-test extraction quality beyond standard formats. The resulting 95.3% average accuracy was computed across all eight scored fields and all 134 resumes weighted equally.